SB6 · STATIC-PLATE

Thick rectangular plate bending — transverse shear, uniform load

Thick-plate transverse shear — production slab on the STRIX engine vs exact FSDT (Mindlin) theory

PASS

error −0.156% · tol 0.5%

§2 TECHNICAL FEATURES TESTED

- **STRIX engine 4-node MITC plate** — production flat-plate slab element
- Elastic membrane-plate section (Mindlin/FSDT plate)
- Transverse-shear correction $\kappa = 5/6$
- Hard simply-supported, uniform pressure
- Thickness sweep $a/t = 50 \to 5$ (6 cases)
- Exact FSDT Navier series (closed form)
- h-refinement convergence to closed form

§3 PROBLEM DESCRIPTION

A **thick** rectangular plate carries a uniform out-of-plane pressure and its center deflection is compared to the **exact first-order shear-deformation (FSDT / Reissner–Mindlin) Navier series**. This is the companion to SB5: where SB5 keeps the plate thin ($a/t = 200$) so transverse shear vanishes, SB6 makes it thick (a/t from 50 down to **5**) so **transverse-shear deflection** becomes a large part of the total (2–17% here) and the Mindlin element must capture it. Following **CSI SAP2000 Verification Example 2-012** (Plate Bending with Shear Deformation; Roark & Young), six configurations are run — a square (1:1) and a 2:1 plate, each at several thicknesses, all **hard simply-supported** under uniform pressure. STRIX recovers the scale-invariant coefficient $\alpha = w \cdot D/(q \cdot a^4)$ and compares it to the series value. The reference uses the **same continuum theory the element discretizes** (FSDT with $\kappa = 5/6$), so the comparison is like-for-like and near-exact; the shear bracket $[1 + D \cdot k^2/(\kappa G h)]$ tends to 1 as the plate thins, recovering the SB5 / Timoshenko value. The plate is a **production flat-plate slab**, meshed and solved on the **STRIX engine** with the same 4-node MITC plate (elastic membrane-plate section) a real STRIX slab uses.

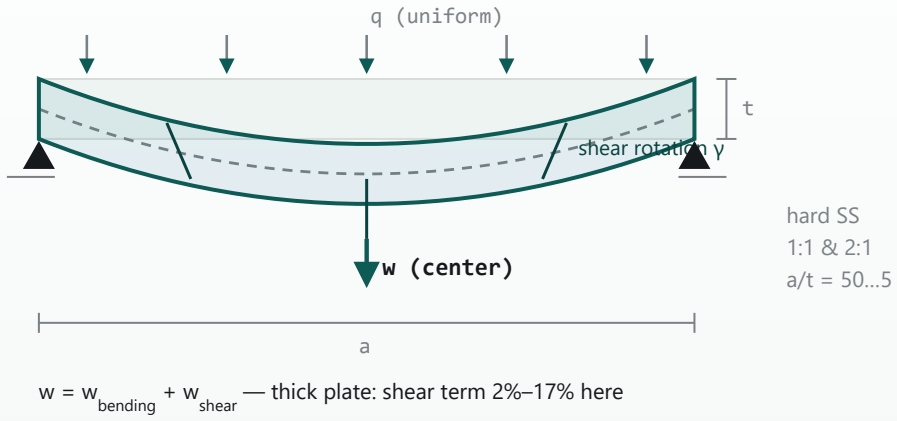


Figure SB6. Thick simply-supported plate (edge section) — the total center deflection w is the bending deflection (dashed mid-surface) plus the extra transverse-shear deflection; the Mindlin cross-section rotates by the shear angle γ. Monitored by the center deflection w.

§4 GEOMETRY, PROPERTIES & LOADING

GEOMETRY		MATERIAL		LOADING & BC	
Short side a	2000 mm	Model	Elastic · Mindlin FSDT	Uniform q	1×10^{-4} N/mm²
Aspects	1:1 & 2:1	E	30000 MPa	Supports	hard SS
Thickness t	40 – 400 mm	ν	0.3	Load	uniform
a / t	50 → 5	κ (shear)	5 / 6	Probe	center defl. w

§5 REFERENCE SOLUTION

α — 1:1, a/t=10: w = α·q·a⁴/D, FSDT κ=5/6 (shear 4.9%) = 0.004273

α — 1:1, a/t=5: w = α·q·a⁴/D, FSDT κ=5/6 (shear 17.2%) = 0.004904

Exact first-order shear-deformation (FSDT / Reissner-Mindlin) Navier series for a hard simply-supported rectangular plate under uniform pressure (Wang relationship, exact for SS plates): $w_{mn} = Q_{mn}/(D\cdot k^4)\cdot[1 + D\cdot k^2/(\kappa G h)]$, κ = 5/6, ν = 0.3. The eight-case-style layout and thick-plate/shear focus follow CSI SAP2000 Verification Example 2-012 (Plate Bending with Shear Deformation; Roark & Young). STRIX recovers the scale-invariant deflection coefficient $\alpha = w\cdot D/(q\cdot a^4)$, a = short side; the shear bracket → 1 as a/t → ∞ (recovers the SB5/Timoshenko thin-plate value) and grows with thickness.

§6 RESULTS COMPARISON

RESPONSE QUANTITY	PROBE	STRIX	REFERENCE	DIFF	VERDICT
Deflection coefficient α	1:1 · a/t=50	0.004068	0.004071	−0.068%	PASS
Deflection coefficient α	1:1 · a/t=20	0.004113	0.004115	−0.061%	PASS
Deflection coefficient α	1:1 · a/t=10	0.004271	0.004273	−0.057%	PASS
Deflection coefficient α	1:1 · a/t=5	0.004903	0.004904	−0.023%	PASS
Deflection coefficient α	2:1 · a/t=10	0.010438	0.010454	−0.156%	PASS
Deflection coefficient α	2:1 · a/t=5	0.011414	0.01143	−0.139%	PASS

§7 CONVERGENCE

MESH	ALPHA	DIFF
8×16	0.010312	−1.362%
16×32	0.010419	−0.337%
24×48	0.010438	−0.156%

For every case the first-order MITC plate approaches the FSDT coefficient `monotonically from below` (a coarse mesh is slightly stiff), reaching the closed form to within 0.16% at the finest mesh (0.07% or less for the square plate). Because the reference is the exact FSDT solution with the element's own shear correction (κ = 5/6), the match is near-exact across the whole thin-to-thick range — the recovered coefficient rises from the thin-plate value at a/t = 50 to a strongly shear-amplified value at a/t = 5, tracking the analytical shear term throughout. §7 shows the governing (worst final) case; all six appear at the finest mesh in §6. The auto rotational stabiliser is stripped.

§8 CONCLUSION

PASS

Across all six configurations — two aspect ratios and thicknesses from `a/t = 50` to a very thick `a/t = 5` — the recovered deflection coefficient matches the exact FSDT (Reissner–Mindlin) Navier solution to a fraction of a percent, converging monotonically. The production slab element on the **STRIX engine** (4-node MITC plate) is verified to reproduce **transverse-shear plate bending** with the correct shear correction (κ = 5/6) over the full thickness range — the thick-plate complement to the thin-plate SB5, together spanning the slab bending regime a real structure spans (flat slabs, transfer plates, mat foundations).

§9 REFERENCES & ANALYSIS FILES

1. Exact first-order shear-deformation (FSDT / Reissner–Mindlin) Navier series for a hard simply-supported plate under uniform load: $w_{mn} = Q_{mn}/(D\cdot k^4)\cdot[1 + D\cdot k^2/(\kappa G h)]$, κ = 5/6, ν = 0.3 (Wang relationship, exact for SS plates); see §5.
2. R. J. Roark & W. C. Young, *Formulas for Stress and Strain* — flat plates with shear deformation; the thick-plate/shear focus follows CSI SAP2000 Software Verification Example 2-012 (Plate Bending with Shear Deformation).
3. STRIX headless harness: `production flat_plate slab → generatePyForLoadCase → STRIX engine (static-single route)`; 4-node MITC plate + elastic membrane-plate section (κ = 5/6), hard simply-support, auto rotational stabiliser removed.

EVIDENCE ARCHIVE `verif-evidence-strix0.134.1-win-x64.zip · SB6/` `sha256 (pending publish)`